

① Determine the following:

i) SWR

ii) Load admittance

iii) Distance b/t load and the first voltage minimum along the transmission line.

iv) For a line with a characteristic impe. of 300Ω and terminated in a load of $175 + j207\Omega$. An electrical signal of 200MHz is transmitted along the line in free space.

iii) Dist. b/t 1st min to load = $0.384\lambda = 0.384(1.5) = 0.576\text{m}$.

$$\lambda = \frac{v}{f} = \frac{3 \times 10^8}{200 \times 10^6} = 1.5 \text{ m}$$

$$Z_0 = 300 \Omega, Z_R = 175 + j207 \Omega$$

$$Z_R = Z_L = 175 + j207 \Omega$$

$$\boxed{Z_r = \frac{Z_R}{Z_0}} = \frac{175 + j207}{300} = 0.5833 + j0.69$$

$$Z_r = 0.58 + j0.69 \rightarrow \text{normalized.}$$

$$Z_r \approx 0.6 + j0.7$$

$$0.7 - j0.8$$

$$S = 2.6$$

$$\frac{1}{S} = \frac{1}{2.6} = 0.38$$

i) $\therefore S = 2.6$ ✓

ii) $Y_r = \frac{1}{Z_r} = 0.71 - j0.84 = 0.7 - j0.8$

$$Y_R = Y_r Z_0 = (0.7 - j0.8) \times 300 = 210 - j240$$

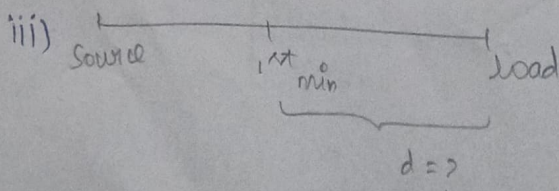
↓

$$\textcircled{x} \quad \boxed{Y_r = \frac{Y_R}{Y_0}} = Y_R Z_0 \quad \text{from graph}$$

$$Y_R = \frac{Y_r}{Z_0} = \frac{0.7 - j0.8}{300} = 0.0023 - j0.0026$$

⇒ $Y_R = \frac{1}{Z_R} = \frac{1}{175 + j207} = 0.0023 - j0.0028$ (verify)

Theoretical calc.



$$d = 0.5\lambda - 0.116\lambda = 0.384\lambda = 0.384(1.5) = 0.576 \text{ m.}$$

2 - V - 3 x 10⁸

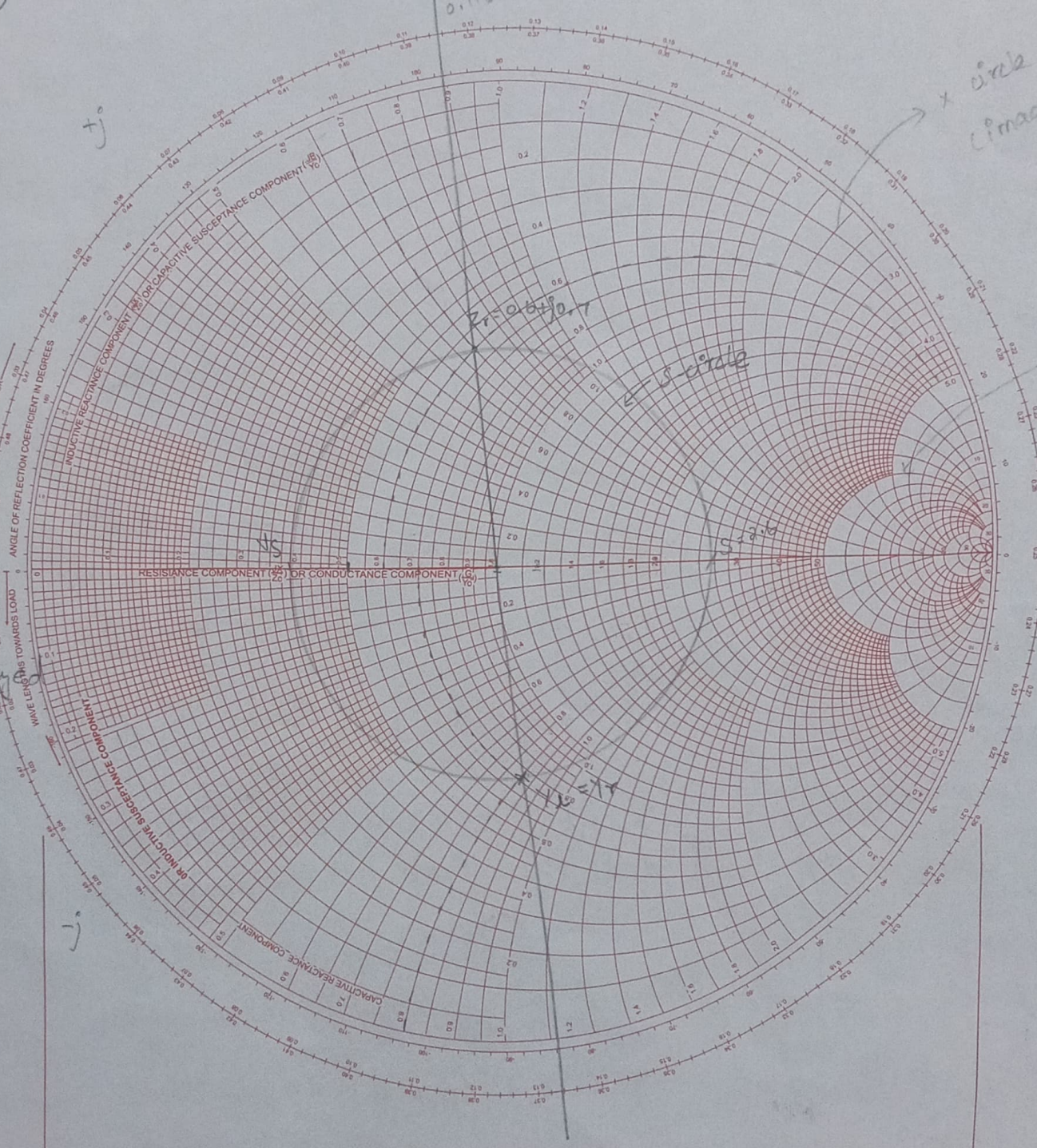
①

0.115λ (or)
0.116λ

+j

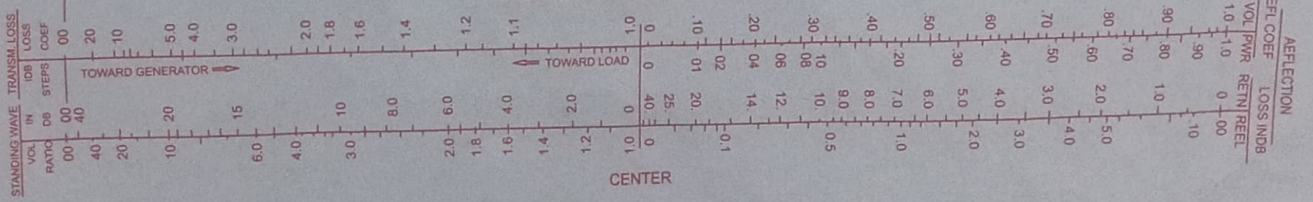
x circle
imaginary

S.C
Z_R = 0
0.5
Z_{min},
normalized
V_{min}



R
G
D
Z_p
Z_m
V_m

RADIALLY SCALED PARAMETERS

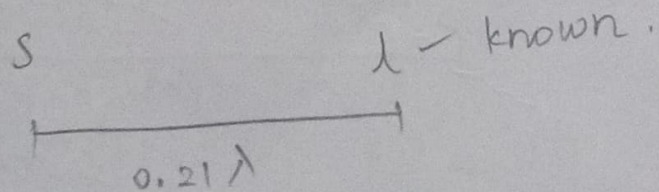


SMITH CHART

$Z_{max} = Z_0 S$

$S = \frac{Z_{max}}{Z_0}$

3. Determine the input impedance and SWR for distance 0.21λ from the load towards generator with $Z_0 = 50\Omega$ and a load impedance is $80 - j30\Omega$.



$$Z_R = 80 - j30$$

$$Z_r = \frac{(80 - j30)}{50} = 1.6 - j0.6$$

$$S = 1.9 \quad \checkmark$$

$$\frac{1}{S} = \frac{1}{1.9} = 0.52$$

$$0.294\lambda + 0.21\lambda = 0.504\lambda$$

$$Z_{IN} = Z_{in} Z_0 = (0.5 + j0.03)(50)$$

$$Z_{IN} = 25 + j1.5 \, \Omega \quad \checkmark \quad (26 + j1.5) \, \Omega$$

4. Determine the input impedance and SWR for a distance 1.25λ long transmission line at the sending end with a characteristic impedance 50Ω and a load impedance $Z_L = 30 + j40$.

$1\lambda + 0.25\lambda \rightarrow 2 \text{ full circle} + 0.25\lambda$

5. on a transmission line a standing wave pattern is observed and VSWR is found to be 3 and it is also noted that two successive voltage minima are 30cm apart. The first voltage

$$S = 1.9 \quad \checkmark$$

$$\frac{1}{S} = \frac{1}{1.9} = 0.52$$

$$0.294\lambda + 0.21\lambda = 0.504\lambda$$

$$Z_{IN} = Z_{in} Z_0 = (0.5 + j0.03)(50)$$

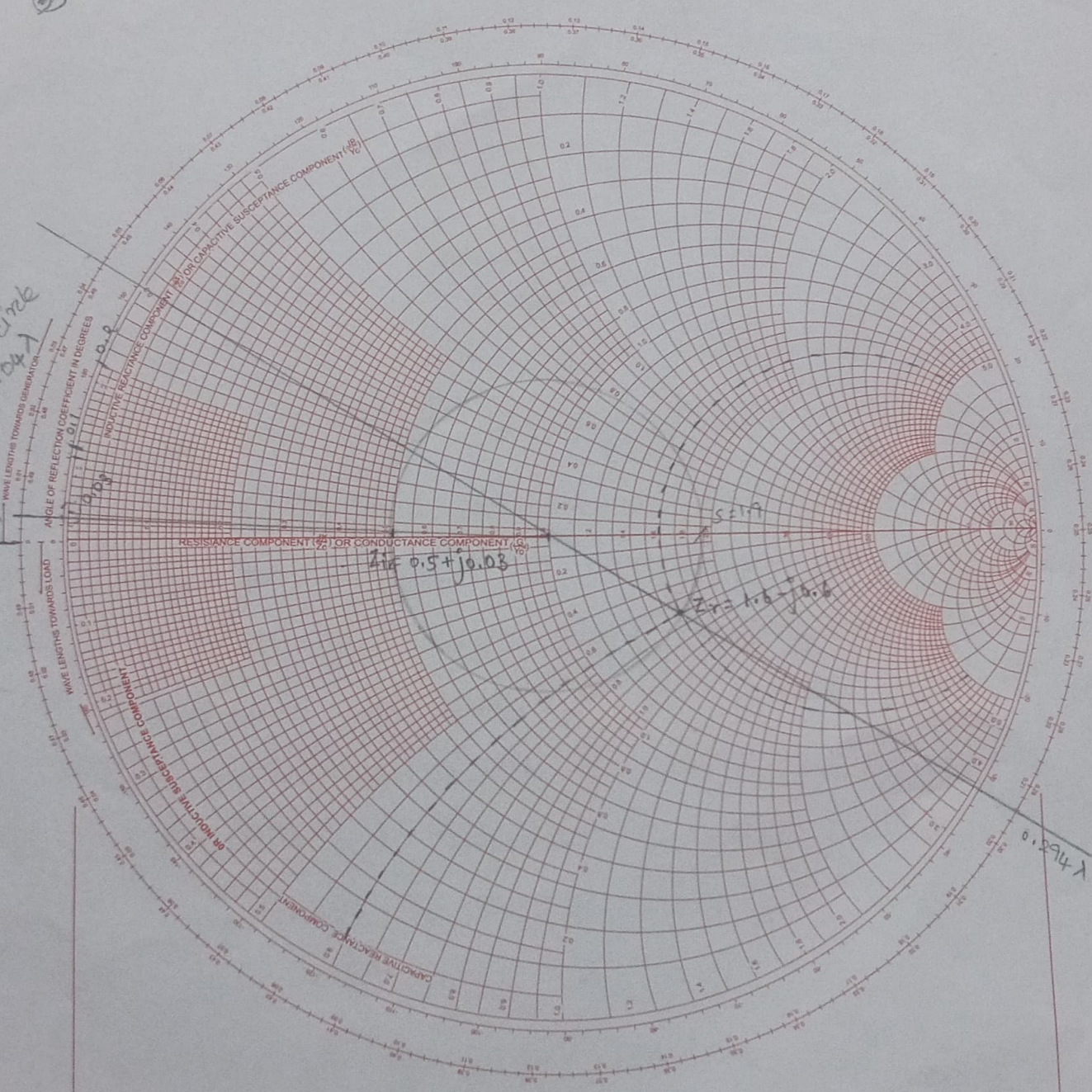
$$Z_{IN} = 25 + j1.5 \quad \Omega, \quad \checkmark$$

$$(26 + j1.5) \Omega.$$

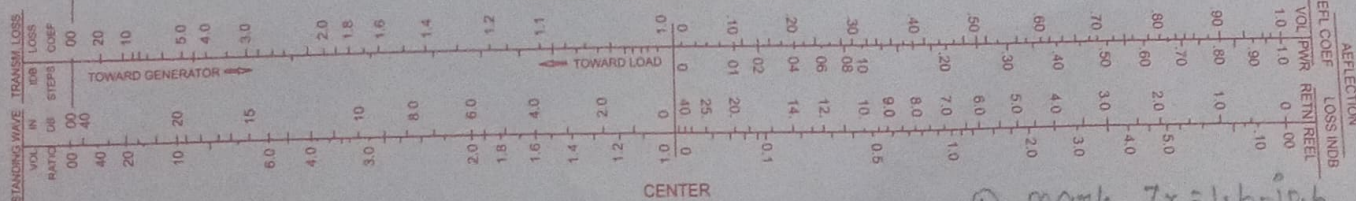
3

1 full circle
to 0.5λ

0.504λ
↑
0.5λ



RADIALLY SCALED PARAMETERS



- ① mark $Z_r = 1.6 - j0.6$
- ② Draw a line from 1 to outer circle via Z_r .

1 complete circle = $\lambda/2 = 0.5\lambda$. **SMITH CHART**

5. on a transmission line a standing wave pattern is observed and VSWR is found to be 3 and it is also noted that two successive voltage minima are 30cm apart. The first voltage minimum is found to be 12cm from the load. The length of the line is 85.2cm and Z_0 is 200Ω . Find

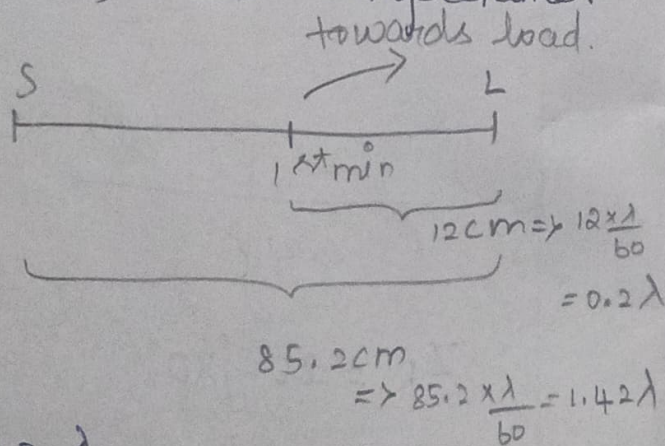
i) sending end impedance ii) load impedance.

soln: Given,

$$S = 3$$

$$\frac{\lambda}{2} = 30\text{cm}$$

$$\lambda = 60\text{cm} \Rightarrow 1\text{cm} = \frac{\lambda}{60}$$



$$i) \quad I_{IN} = I_{in} Z_0 = (2.3 + j1.3) 200 = 460 + j260 \, \Omega$$

$$ii) \quad I_L = I_L Z_0 = (1.7 - j1.35) 200 = 340 - j270 \, \Omega$$

S

5

200

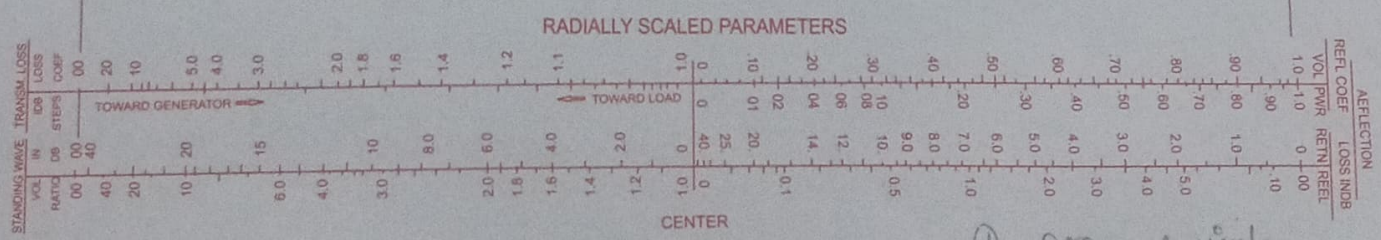
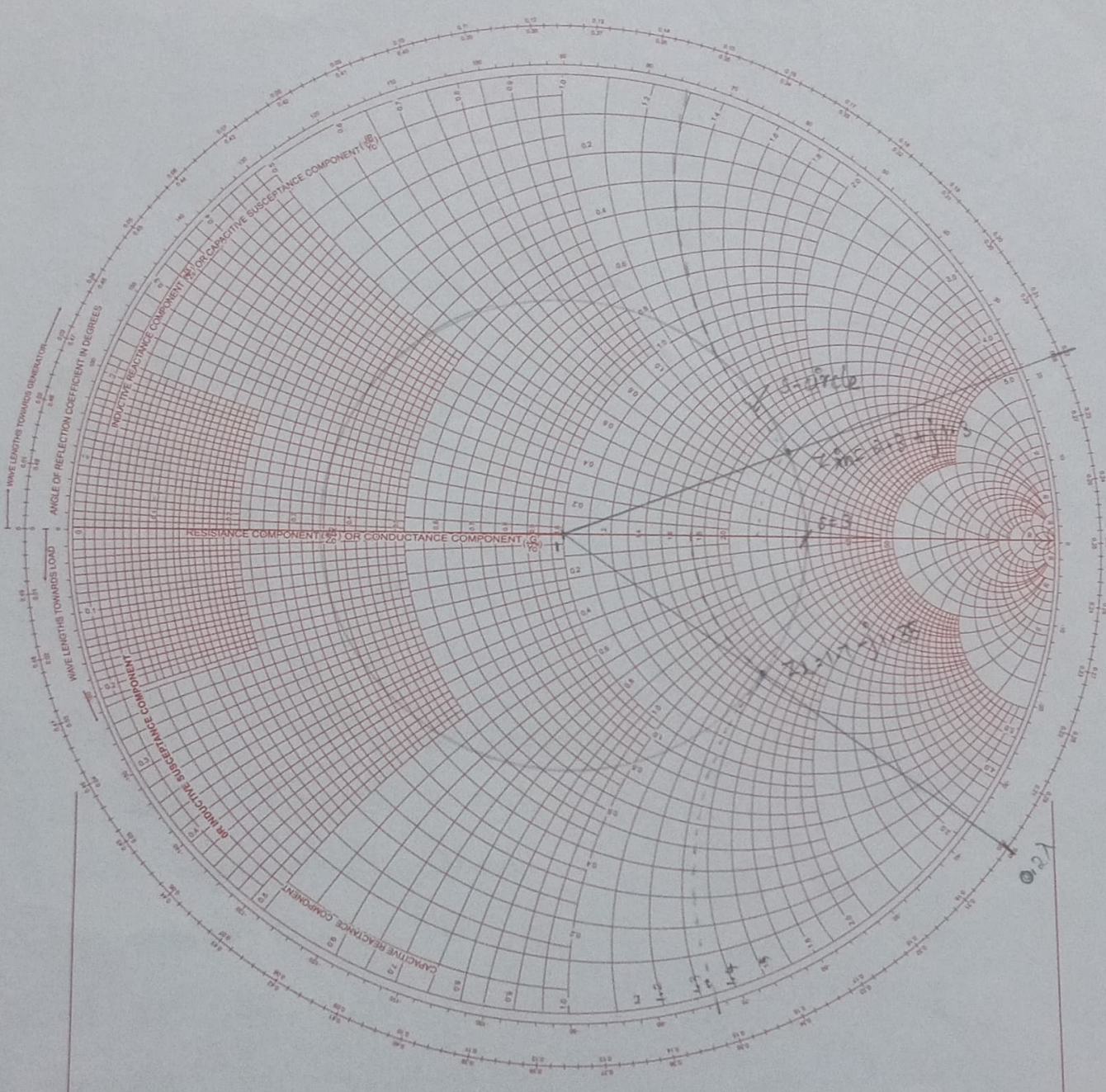
20

D.

Tr

V_{min}

V_{max}



SMITH CHART

$$0.42\lambda - 0.2\lambda = 0.22\lambda$$

- ① Draw s-circle $s=3$.
- ② mark V_{min} towards load (0.2λ) .

6. VSWR on a lossless line is found to be 5 and wavelength of line is 80 cm. First voltage min. is observed 15 cm from load. length of line is 160 cm and characteristic impedance is 300Ω . Using smith chart, determine I_L , sending end impedance.

$$S = 5$$

$$Z_0 = 300 \Omega$$

$$\lambda = 80 \text{ cm.}$$

$$\frac{\lambda}{80} = 1 \text{ cm}$$

$$15 \times \frac{\lambda}{80} = 0.1875 \lambda$$

$$160 \times \frac{\lambda}{80} = 2 \lambda$$

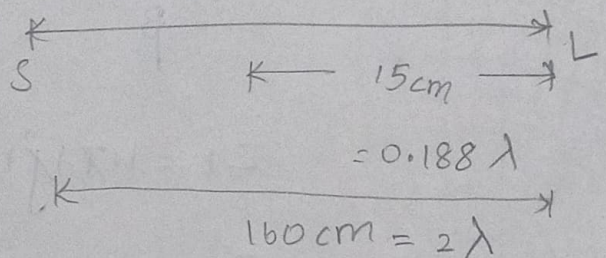
$$Z_L = 1.1 - j1.9$$

$$Z_L = Z_L Z_0 = (1.1 - j1.9) 300$$

$$Z_L = 330 - j570 \Omega \checkmark$$

$$Z_S = 1.1 - j1.9$$

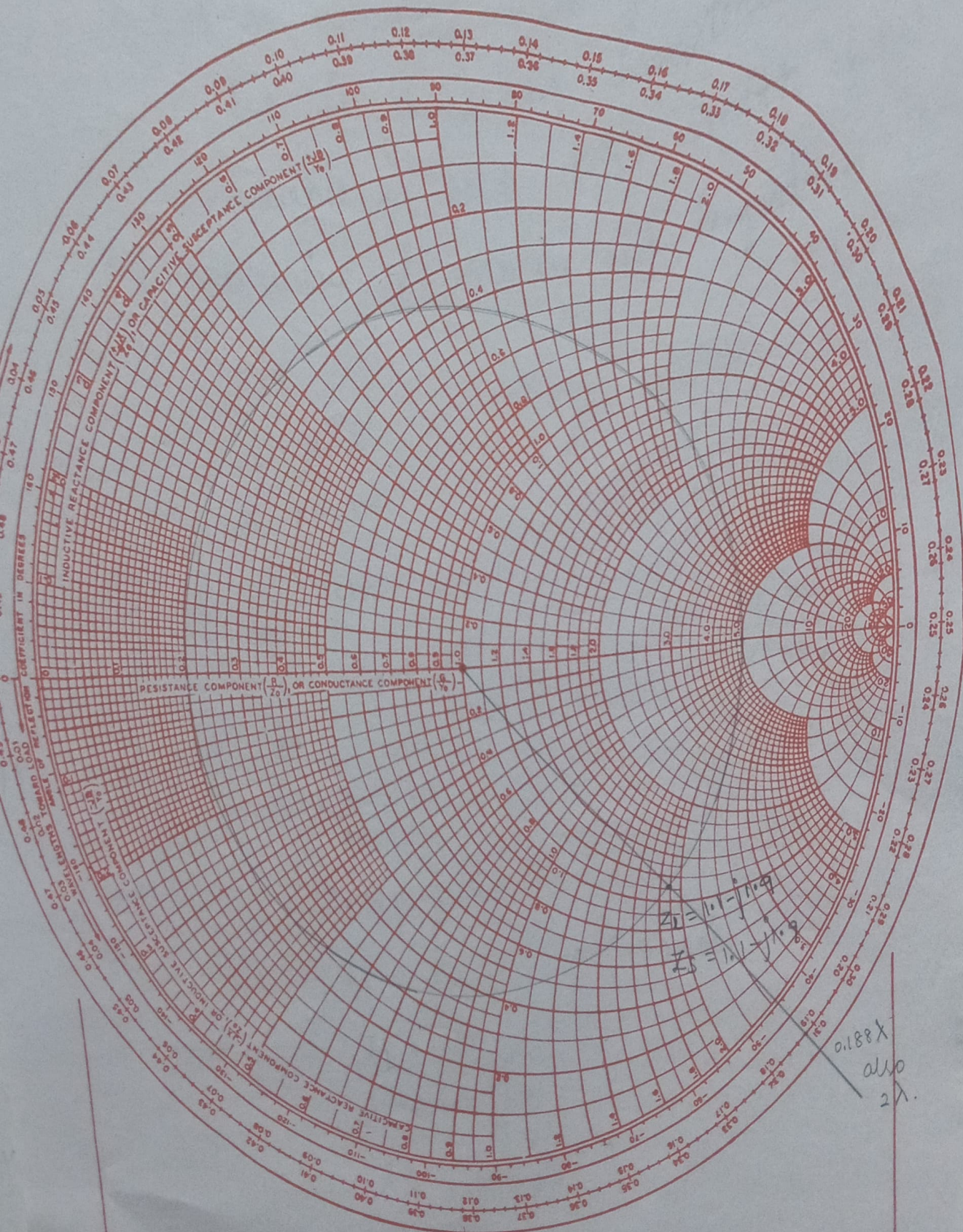
$$Z_S = 330 - j570 \Omega \checkmark$$



$$k = \frac{Z_R - Z_0}{Z_R + Z_0}$$

SMITH-CHART

IMPEDANCE OR ADMITTANCE COORDINATES

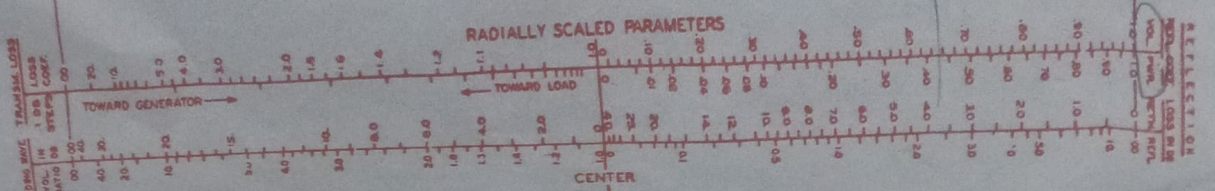


$$Z_L = 1.0 - j1.0$$

$$Y_L = 1.0 + j1.0$$

0.188λ
also
2λ

RADIALLY SCALED PARAMETERS



7. A lossless line of characteristic impedance $100\ \Omega$ is terminated unknown load impedance producing SWR of value 3. The operating freq. is 300 MHz . Determine unknown load impedance if first voltage maxima occurs

at 320 cm from load.

soln:

$$Z_0 = 100 \Omega$$

$$S = 3$$

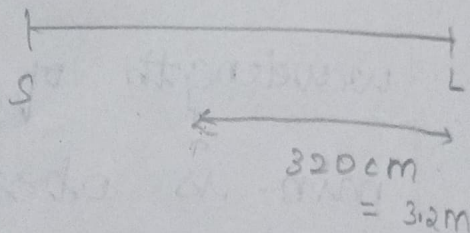
$$f = 300 \text{ MHz}$$

$$Z_L = ?$$

$$\lambda = \frac{2\pi}{\beta} = \frac{v}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}$$

$$Z_L = 100 + j138 \quad 0.36 + j0.29$$

$$Z_L = Z_L \cdot Z_0 = 100 + j138 \Omega = 36 + j29 \Omega$$



$$\lambda = 1 \text{ m}$$

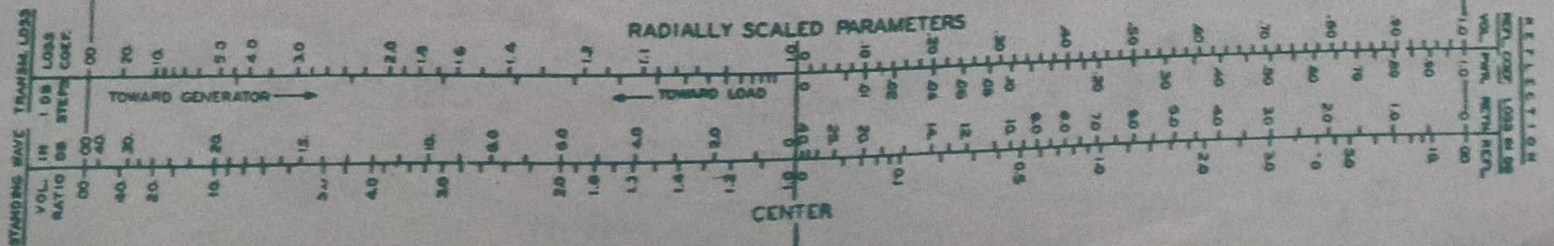
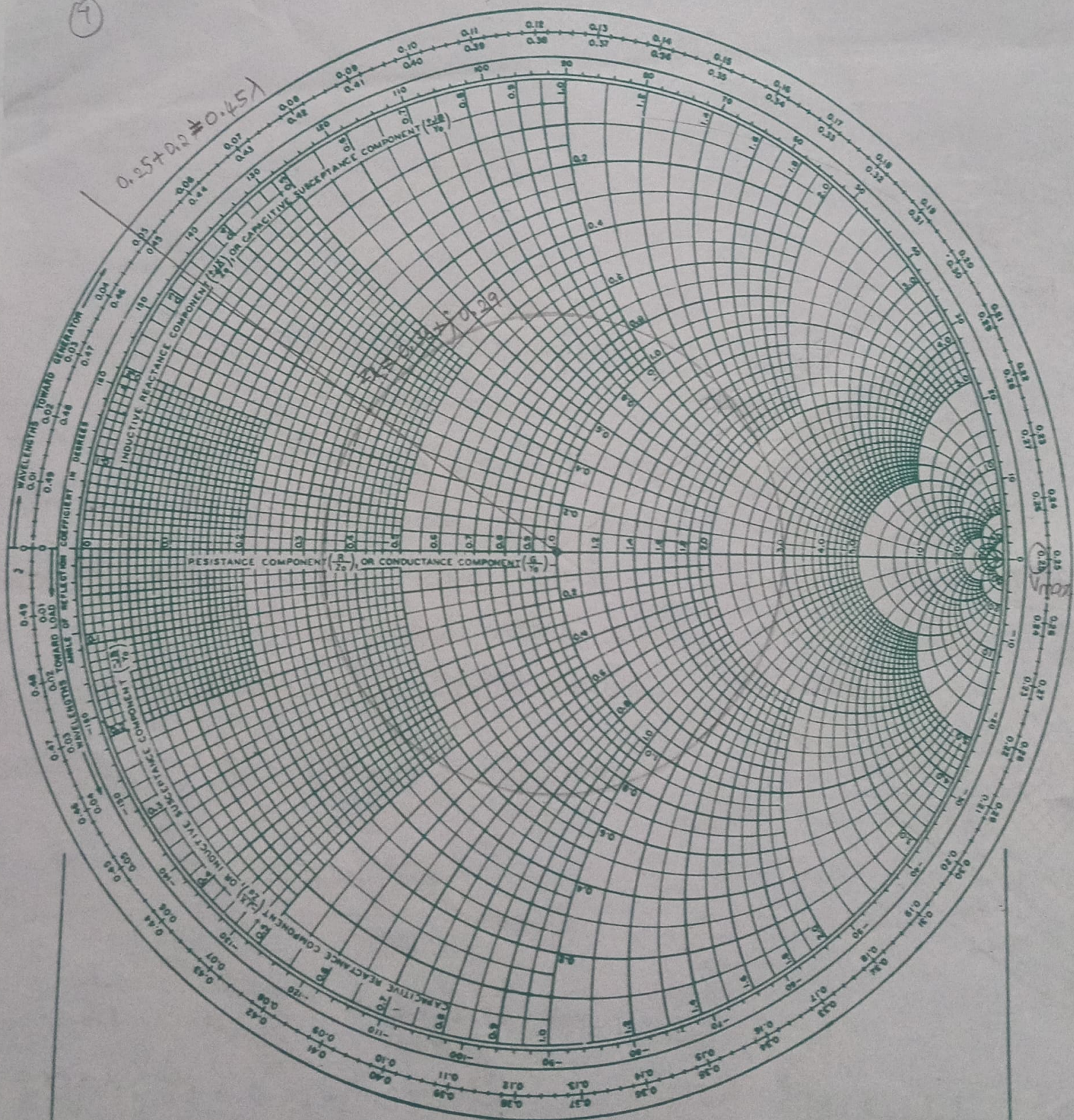
$$1 \text{ m} = \lambda$$

$$3.2 \text{ m} = 3.2 \lambda$$

$$c = 3 \times 10^8 \text{ m/s}$$

	SMITH-CHART	

IMPEDANCE OR ADMITTANCE COORDINATES



Single stub matching

8. 300Ω transmission line is connected to a load impedance of $450 - j600\Omega$ at a freq. of 10MHz . Find stub location and stub length of a short-circuited stub required to match the line using Smith chart.

$$Z_0 = 300\Omega, \quad Z_R = Z_L = 450 - j600\Omega, \quad f = 10\text{MHz}.$$

$$Z_r = Z_L = \frac{Z_R}{Z_0} = \frac{450 - j600}{300} = 1.5 - j2$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{1.5 - j2} = 0.24 + j0.32 \quad (\text{Theoretically})$$

from graph, $Y_L = 0.24 + j0.33$

$$Y_L = Y_L Y_0 = \frac{Y_L}{Z_0} = 0.0008 + j0.001 \quad \checkmark$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10 \times 10^6} = 30 \text{ m}.$$

$$\lambda_s = 3.9 \text{ m}$$

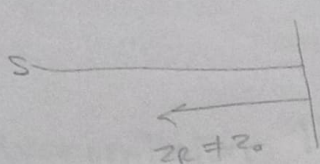
(stub location)

$$\lambda_t = 2.61 \text{ m}$$

(stub length)

Unit - II

Measurement of unknown load impedance



unknown Z_L

$$V_{min} = \frac{1}{2.5} = 0.4$$

$$K = \frac{Z_R - Z_0}{Z_R + Z_0} \quad ; \quad S = \frac{V_{max}}{V_{min}} \quad ; \quad S = \frac{1 + |K|}{1 - |K|}$$

$$S = 1$$

$$Z_L = Z_{min} = \frac{Z_0}{S}$$

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan(\beta l)}{Z_0 + j Z_R \tan(\beta l)} \right]$$

$$\beta = \frac{2\pi}{\lambda}$$

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan\left(\frac{2\pi l}{\lambda}\right)}{Z_0 + j Z_R \tan\left(\frac{2\pi l}{\lambda}\right)} \right]$$

$$\frac{Z_0}{S} = Z_0 \left[\frac{Z_R + j Z_0 \tan\left(\frac{2\pi l'}{\lambda}\right)}{Z_0 + j Z_R \tan\left(\frac{2\pi l'}{\lambda}\right)} \right]$$

$$Z_0 + j Z_R \tan\left(\frac{2\pi l'}{\lambda}\right) = S \left[Z_R + j Z_0 \tan\left(\frac{2\pi l'}{\lambda}\right) \right]$$

$$Z_0 - j S Z_0 \tan\left(\frac{2\pi l'}{\lambda}\right) = S Z_R - j Z_R \tan\left(\frac{2\pi l'}{\lambda}\right)$$

$$Z_0 \left[1 - j S \tan\left(\frac{2\pi l'}{\lambda}\right) \right] = Z_R \left[S - j \tan\left(\frac{2\pi l'}{\lambda}\right) \right]$$

$$Z_R = Z_0 \left[\frac{1 - j S \tan\left(\frac{2\pi l'}{\lambda}\right)}{S - j \tan\left(\frac{2\pi l'}{\lambda}\right)} \right]$$

Open and short Circuited Lines

$\Rightarrow$ open circuited line is defined as a transmission line whose far end (terminating end) is open.

$\Rightarrow$ short-circuited line is defined as a transmission line whose far end is shorted.

Standing waves in open and short-circuited lines.

a) open circuited lines:

$\Rightarrow$ At open end, there exists a maximum voltage and minimum current (nearly zero), impedance $\rightarrow \infty$

$\Rightarrow$ A quarter-wavelength from the open end, incident wave will be 90° earlier and reflected wave 90° later, 180° out of phase, so voltage at this point is zero.

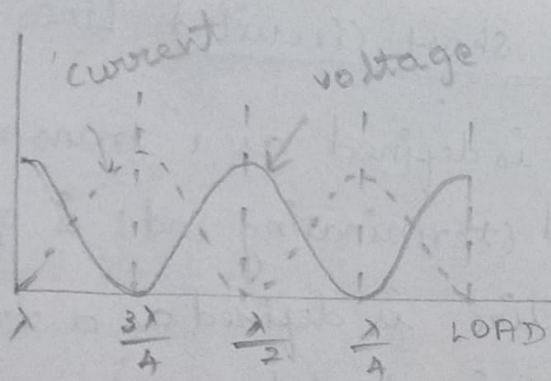
$\Rightarrow$ The standing wave pattern is repeated every half-wave lengths, i.e., maxima are spaced half-wavelength apart on transmission line and minima are also spaced half a wavelength apart.

$\Rightarrow$ Distance b/w a maximum and minimum is a quarter wavelength.

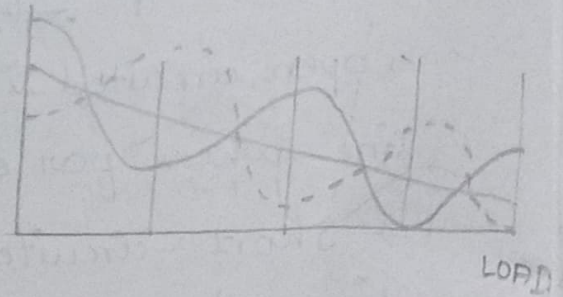
$\Rightarrow$ Current maximum occurs at a point of voltage minimum and vice-versa.

$\Rightarrow$ In high freq. lossless line, values of different maximum are equal.

$\Rightarrow$ In a lossy line, these go on decreasing due to attenuation of the line.



Lossless line



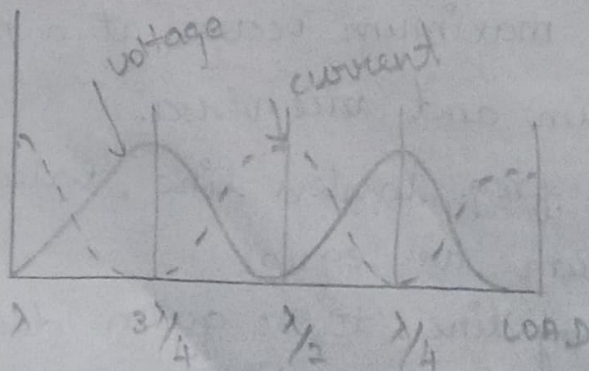
Lossy line

b) Short-circuited lines:

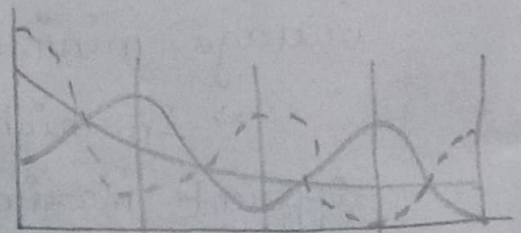
$\Rightarrow$ At short-circuited termination, current is maximum, voltage is zero, impedance is zero.

$\Rightarrow$ Standing waves thus has a node or minimum at short-circuited end and at every half wavelength from the end.

$\Rightarrow$ Voltage on the line goes through minima at distances from the load that are even multiples of quarter wavelength and through maxima at a distance that are odd multiple of quarter wavelength.



Lossless line



Lossy line

Input impedance of open and short circuited lines

Low freq: ($l \ll \lambda$), $v, i \Rightarrow$ almost constant throughout line.

$$Z_{oc} = Z_0 \coth \gamma l, \quad Z_{sc} = Z_0 \tanh \gamma l$$

For an infinite length of line $l = \infty$,

$$\tanh \gamma l = 1, \quad \coth \gamma l = 1$$

$$\tanh \gamma l = \frac{\sinh \gamma l}{\cosh \gamma l}$$

$$= \frac{(e^{\gamma l} - e^{-\gamma l})/2}{(e^{\gamma l} + e^{-\gamma l})/2}$$

$$= \frac{e^{\gamma l} - e^{-\gamma l}}{e^{\gamma l} + e^{-\gamma l}}$$

$$= \frac{e^{\gamma l} [1 - e^{-2\gamma l}]}{e^{\gamma l} [1 + e^{-2\gamma l}]}$$

$$= \frac{1 - e^{-2\gamma l}}{1 + e^{-2\gamma l}}$$

$$= \frac{1 - e^{-\infty}}{1 + e^{-\infty}} = \frac{1 - 0}{1 + 0} = 1$$

then $Z_{oc} = Z_{sc} = Z_0$

$\therefore$ Input impedance of an infinite line is its characteristic impedance.

$\Rightarrow I = 0$, voltage reaches the ^{open ckt} open end, so line behaves like a small capacitor.

$\Rightarrow$ Reflection effects are negligible.

^{short ckt}

$\Rightarrow I = \text{max}$, $V = 0$, current flows easily, almost no resistance ($Z_R = 0$). Behaves like a small inductor. There is a path for current flow.

At High Freq: When wave reaches open end, it can't continue, it reflects back towards the source. $\lambda = \lambda$, $V, I \Rightarrow$ vary; reflection occurs. Standing waves are produced.
 $Z_{oc} = jZ_0 \cot \beta l$, $Z_{sc} = j \tan \beta l (Z_0)$ $k=1$

$\Rightarrow$ Input impedance of an open and short-circuited lossless line is a pure reactance. (inductive or capacitive).

Equivalent inductance impedance is obtained

$$j\omega L_{eq} = jZ_0 \tan \beta l \quad (0 \leq l \leq \lambda/4)$$

$$L_{eq} = \frac{Z_0}{\omega} \tan \beta l$$

Equivalent capacitive impedance is given by

$$-\frac{j}{\omega C_{eq}} = jZ_0 \tan \beta l \quad \left[\frac{\lambda}{4} \leq l \leq \frac{\lambda}{2} \right]$$

$$C_{eq} = \frac{1}{\omega Z_0 \tan \beta l}$$

$\Rightarrow$ Considering the variation of Z_{oc} only, parallel resonance occurs when βl is even multiple of $\frac{\pi}{2}$ and series resonance occurs when βl is odd multiple of $\frac{\pi}{2}$. The graph from $\beta l = 0$ to $\beta l = \pi$ has the general form as the reactance curve of series lumped LC circuit, when freq. is varied.

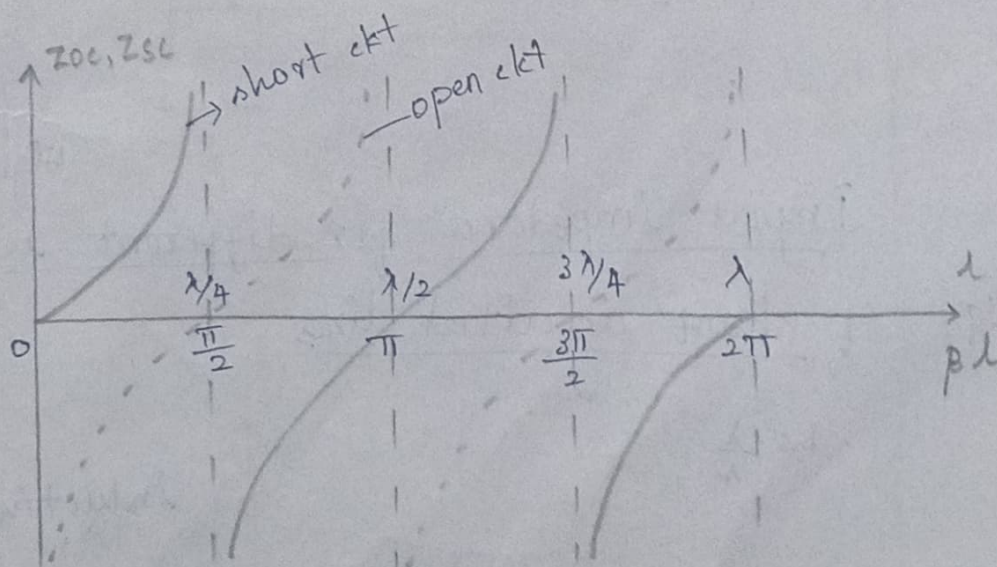
$\Rightarrow$ Considering the variation of Z_{sc} only, parallel resonance or anti-resonance with theoretically infinite impedance occurs when βl is odd multiple of $\frac{\pi}{2}$ and series resonance or resonance with theoretically zero impedance occurs when

βl is even multiple of $\frac{\pi}{2}$. Graph from $\beta l = 0$ to $\beta l = \pi$ has the same general form as the reactance curve of a parallel lumped LC circuits when freq. is varied.

$$\beta = \frac{2\pi}{\lambda}$$

$$Z_{oc} = -jZ_0 \cot\left(\frac{2\pi}{\lambda} l\right), \quad Z_{sc} = jZ_0 \tan\left(\frac{2\pi}{\lambda} l\right)$$

βl	in terms of λ	Z_{oc}	Z_{sc}
0	0	$-\infty$	0
$\frac{\pi}{4}$	$\frac{\lambda}{8}$	$-jZ_0$	jZ_0
$\frac{\pi}{2}$	$\frac{\lambda}{4}$	0	∞
$\frac{3\pi}{4}$	$\frac{3\lambda}{8}$	jZ_0	$-jZ_0$
π	$\frac{\lambda}{2}$	∞	0
$\vdots$	$\vdots$		
$\frac{3\pi}{2}$	$\frac{3\lambda}{4}$	0	∞
2π	λ	∞	0



$\Rightarrow$ length of line in terms of λ
electrical length of line βl in terms of π .

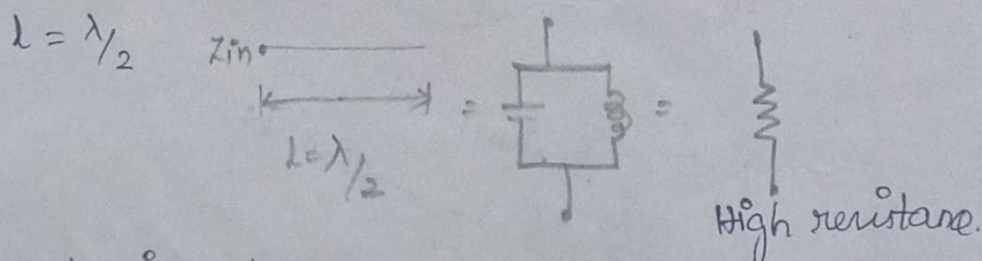
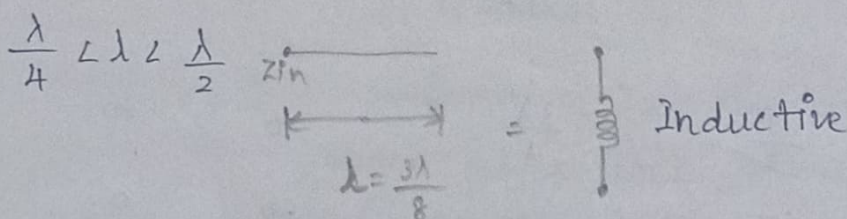
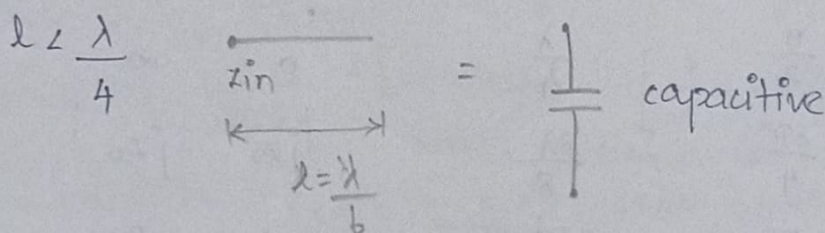
$\Rightarrow$ Above horizontal line, Z_{oc} and Z_{sc} will have positive reactance i.e. inductive.

$\Rightarrow$ Below horizontal line, -ve reactive i.e. capacitive.

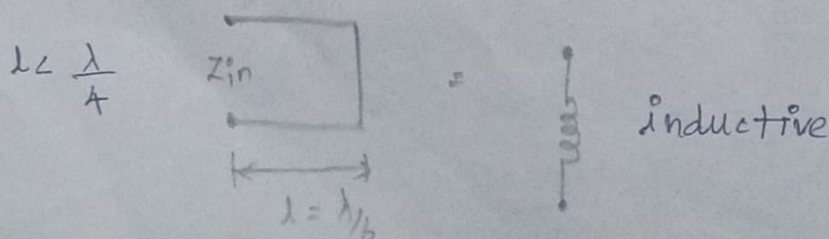
$l < \frac{\lambda}{4} \Rightarrow$ value of $Z_{sc} \rightarrow +ve$ (inductive reactance)
value of $Z_{oc} \rightarrow -ve$ (capacitive reactance).

$\frac{\lambda}{4} < l < \frac{\lambda}{2} \Rightarrow$ reactance of $Z_{sc} \rightarrow$ capacitive
reactance of $Z_{oc} \rightarrow$ inductive.

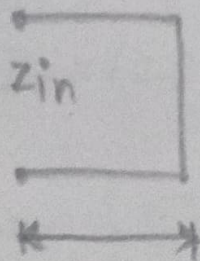
Input impedance for different wavelengths of open circuited lines:



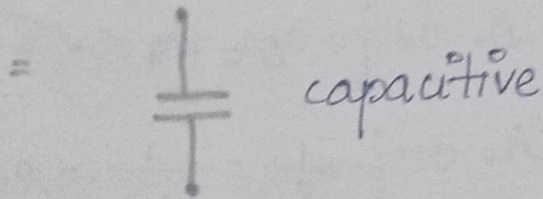
Input impedance for different wavelengths of short circuited lines:



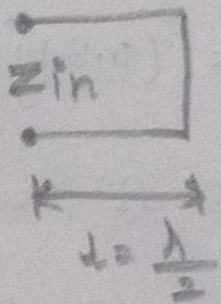
$$\frac{\lambda}{4} < l < \frac{\lambda}{2}$$



$$l = \frac{3\lambda}{8}$$



$$l = \frac{\lambda}{2}$$



$$l = \frac{\lambda}{2}$$



Small resistance.

Stub length & Stub location:

$$Z_S = Z_0 \left[\frac{Z_R + Z_0 \tanh(\beta l)}{Z_0 + Z_R \tanh(\beta l)} \right]$$

$$Y_S = Y_0 \left[\frac{Y_R + Y_0 \tanh(\beta l)}{Y_0 + Y_R \tanh(\beta l)} \right]$$

$$\frac{Y_S}{Y_0} = \frac{\frac{Y_R}{Y_0} + \tanh(\beta l)}{1 + \frac{Y_R}{Y_0} \tanh(\beta l)}$$

$$Y_{in} = \frac{Y_r + \tanh(\beta l)}{1 + Y_r \tanh(\beta l)}$$

$$\alpha = 0, \quad \beta = \alpha + j\beta = j\beta$$

$$Y_{in} = \frac{Y_r + j \tan(\beta l)}{1 + j Y_r \tan(\beta l)}$$

$$Y_S = Y_0 = Y_R$$

$$\frac{Y_S}{Y_0} = Y_{in} = 1$$

$$Y_{in} = \frac{Y_r + j \tan(\beta l)}{1 + j Y_r \tan(\beta l)} \times \frac{1 - j Y_r \tan(\beta l)}{1 - j Y_r \tan(\beta l)}$$

$$Y_{in} = \frac{Y_r(1 + \tan^2 \beta l) + j(1 - Y_r^2) \tan \beta l}{1 + Y_r^2 \tan^2 \beta l}$$

To find stub location: Equate real terms.

$$1 = \frac{Y_r(1 + \tan^2 \beta l)}{1 + Y_r^2 \tan^2 \beta l}$$

$$1 + Y_r^2 \tan^2 \beta l = Y_r(1 + \tan^2 \beta l)$$

$$\gamma_r + \tan^2 \beta l (1 - \gamma_r) = 1 - \gamma_r$$

$$\tan^2 \beta l = \frac{1}{\gamma_r} = \frac{\gamma_0}{\gamma_R}$$

$$\tan \beta l = \sqrt{\frac{\gamma_0}{\gamma_R}}$$

$$\beta l = \tan^{-1} \sqrt{\frac{\gamma_0}{\gamma_R}}$$

$$\left[\beta = \frac{2\pi}{\lambda} \right]$$

$$L_s = \frac{\lambda}{2\pi} \tan^{-1} \left[\sqrt{\frac{\gamma_0}{\gamma_R}} \right]$$

$$L_s = \frac{\lambda}{2\pi} \tan^{-1} \left[\sqrt{\frac{Z_R}{Z_0}} \right] \checkmark$$

To find stub length:

to eliminate the imaginary, we short circuit

$$\frac{S_s}{\gamma_0} = \frac{[1 - \gamma_r^2] \tan(\beta l)}{1 + \gamma_r^2 \tan(\beta l)}$$

$$\tan(\beta l) = \sqrt{\frac{\gamma_0}{\gamma_R}}$$

$$\frac{S_s}{\gamma_0} = \frac{[1 - \gamma_r^2] \sqrt{\frac{\gamma_0}{\gamma_R}}}{1 + \gamma_r^2 \times \frac{\gamma_0}{\gamma_R}}$$

$$= \frac{\left[1 - \frac{\gamma_R^2}{\gamma_0^2} \right] \sqrt{\frac{\gamma_0}{\gamma_R}}}{1 + \frac{\gamma_R^2}{\gamma_0} \times \frac{\gamma_0}{\gamma_R}}$$

$$= \frac{\left[1 - \frac{\gamma_R^2}{\gamma_0^2} \right] \sqrt{\frac{\gamma_0}{\gamma_R}}}{1 + \frac{\gamma_R}{\gamma_0}}$$

$$= \frac{\left[1 - \frac{\gamma_R^2}{\gamma_0^2} \right] \sqrt{\frac{\gamma_0}{\gamma_R}}}{1 + \frac{\gamma_R}{\gamma_0}}$$

$$1 + \frac{\gamma_R}{\gamma_0}$$

$$= \left[1 + \frac{Y_R}{Y_0} \right] \left[1 - \frac{Y_R}{Y_0} \right] \sqrt{\frac{Y_0}{Y_R}}$$

$$\frac{S_s}{Y_0} = \frac{1 + \frac{Y_R}{Y_0}}{\left[1 - \frac{Y_R}{Y_0} \right] \sqrt{\frac{Y_0}{Y_R}}}$$

$$= \frac{Y_0 - Y_R}{Y_0} \sqrt{\frac{Y_0}{Y_R}}$$

$$= \frac{Y_0 - Y_R}{\sqrt{Y_0 Y_R}} \frac{\sqrt{Y_0}}{\sqrt{Y_R}}$$

$$= \frac{Y_0 - Y_R}{\sqrt{Y_0 Y_R}}$$

$$= \left[\frac{1}{Z_0} - \frac{1}{Z_R} \right] \sqrt{Z_0 Z_R}$$

$$= \left[\frac{Z_R - Z_0}{Z_0 Z_R} \right] \sqrt{Z_R Z_0}$$

$$\frac{S_s}{Y_0} = \frac{Z_R - Z_0}{\sqrt{Z_0 Z_R}}$$

$$\cot(\beta l_t) = \frac{Z_R - Z_0}{\sqrt{Z_R Z_0}}$$

$$\tan(\beta l_t) = \frac{\sqrt{Z_0 Z_R}}{Z_R - Z_0}$$

$$\beta l_t = \tan^{-1} \left[\frac{\sqrt{Z_0 Z_R}}{Z_R - Z_0} \right]$$

$$l_t = \frac{\lambda}{2\pi} \tan^{-1} \left[\frac{\sqrt{Z_0 Z_R}}{Z_R - Z_0} \right]$$

$$Z_{SC} = j Z_0 \tan \beta l$$

$$Z_{SC} = j \frac{1}{Y_0} \frac{1}{\cot \beta l}$$

$$\cot \beta l = j \frac{1}{Y_0 Z_{SC}}$$

$$\cot \beta l = \frac{S}{Y_0}$$